MA416

Lecture 03 – September 9, 2025
Monte-Carlo (MC) Simulations

Dr. Matthew N. Moore

Review of Some R Functions We Have Used

Standard-Normal (Z) Distributions

- qnorm(A, lower.tail=F) … calculates a critical value associated to area 'A' in the right tail
- qnorm(A, lower.tail=T) … calculates a critical value associated to area 'A' in the left tail
- pnorm(q, lower.tail=F) … calculates the area above 'q' in the right tail
- pnorm(q, lower.tail=T) … calculates the area below 'q' in the lower tail

T Distributions

- qt(A, v, lower.tail=F) … calculates a critical value associated to area 'A' in the right tail with v degrees of freedom
- qt(A, v, lower.tail=T) … calculates a critical value associated to area 'A' in the left tail with v degrees of freedom
- pt(q, v, lower.tail=F) … calculates the area above 'q' in the right tail with v degrees of freedom
- pt(q, v, lower.tail=T) … calculates the area below 'q' in the left tail with v degrees of freedom

F Distributions

- qf(A, v1, v2, lower.tail=F) … calculates a critical value associated to area 'A' in the right tail with (v1, v2) d.o.f.
- qf(A, v1, v2, lower.tail=T) … calculates a critical value associated to area 'A' in the left tail with (v1, v2) d.o.f.
- pf(q, v1, v2, lower.tail=F) … calculates the area above 'q' in the right tail with (v1, v2) d.o.f.
- pf(q, v1 , v2, lower.tail=T) … calculates the area below 'q' in the left tail with (v1, v2) d.o.f.

χ^2 Distributions

Same syntax as T, but qchisq(…) and pchisq(…) instead of qt(…) and pt(…)

Assumptions Matter…

For $H_0: \mu = \mu_0$, we required that the observations are independent and that the sample comes from a normal distribution.

This allowed for $t_{stat} \sim T_{n-1}$ (if σ unknown) and $z_{stat} \sim Z$ (if σ known).

Central limit theorem allows for approximations to T_{n-1} and/or Z for "appropriate" sample size magnitudes.

Q) Can we use other test statistics to assess claims aside from the "classic" ones?

Example: $H_0: \mu = \mu_0$ with $\bar{X}$ as an estimate for μ, with the assumption that $X_k \sim^{iid} N(\mu, \sigma^2)$

Define:

$$\star_{stat} := \frac{\frac{|\bar{x}| + 1}{|\mu_0| + 1} + 3}{\frac{|\bar{x}| + 1}{|\mu_0| + 1} + 2}$$

Assumptions Matter…

Example: $H_0: \mu = \mu_0$ with $\bar{X}$ as an estimate for μ, with the assumption that $X_k \sim^{iid} N(\mu, \sigma^2)$

$$\star_{stat} := \frac{\frac{|\bar{x}|+1}{|\mu_0|+1} + 3}{\frac{|\bar{x}|+1}{|\mu_0|+1} + 2}$$

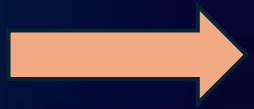
If our goal is to **not reject** H_0, we want $\bar{x} = \mu_0$, meaning that we want $\star_{stat} = \frac{4}{3}$.

If our goal is to **reject** H_0, we want $\bar{x} \neq \mu_0$, meaning that we want $\star_{stat} \ll \frac{4}{3}$ or $\star_{stat} \gg \frac{4}{3}$

```
# claim: mu0 = 140
set.seed(123)
s = rnorm(26, 123, 3.4)

mystar = function(s, mu0){
  top = (abs(mean(s))+1)/(abs(mu0)+1) + 3
  bot = (abs(mean(s))+1)/(abs(mu0)+1) + 2
  return(top/bot)
}

mystar(s, 140)
```



```
> mystar(s, 140)
[1] 1.347573
```

Q) Is this significantly different from $\frac{4}{3}$ to statistically reject H_0?

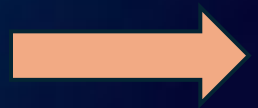
When Parameters are Unknown

Example: $H_0: \mu = \mu_0$ with $\bar{X}$ as an estimate for μ, with the assumption that $X_k \sim^{iid} N(\mu, \sigma^2)$

Note here that we only assume that $X_k \sim$ Normal, but we don't know the parameters of the distribution; so, standardization of our same will "remove" the mean and variance parameter influence from our data set (why?).

```
# claim: mu0 = 140
set.seed(123)
s = rnorm(26, 123, 3.4)
zs = (s - mean(s))/sd(s)
mystar = function(s, mu0){
  top = (abs(mean(s))+1)/(abs(mu0)+1) + 3
  bot = (abs(mean(s))+1)/(abs(mu0)+1) + 2
  return(top/bot)
}
zmu0 = (140 - mean(s))/sd(s)

mystar(s, 140)
mystar(zs, zmu0)
```



```
> mystar(s, 140)
[1] 1.347573
> mystar(zs, zmu0)
[1] 1.462614
```

Monte-Carlo Simulations

```
> mystar(zs, zmu0)
[1] 1.462614
```

* Monte-Carlo simulations samples from the “null” (hypothesized) distribution, and looks at all samples and their associated test statistics to see what (almost) everyone looks like, in order to construct an empirical distribution of the associated test statistic which allows for a set of comparison values for our acquired statistic.

```
set.seed(12121)
mcstarstats = c()
nmc = 100000

for(k in 1:nmc){
  mcs = rnorm(26, 0, 1)
  mcstar = mystar(mcs, zmu0)

  mcstarstats = c(mcstarstats, mcstar)
}
# Run-time approximately 20 seconds

hist(mcstarstats, breaks=100, col='lightgreen', xlim=c(1.435, 1.465))
```

Monte-Carlo Simulations

```
> mystar(zS, zmu0)
[1] 1.462614
```

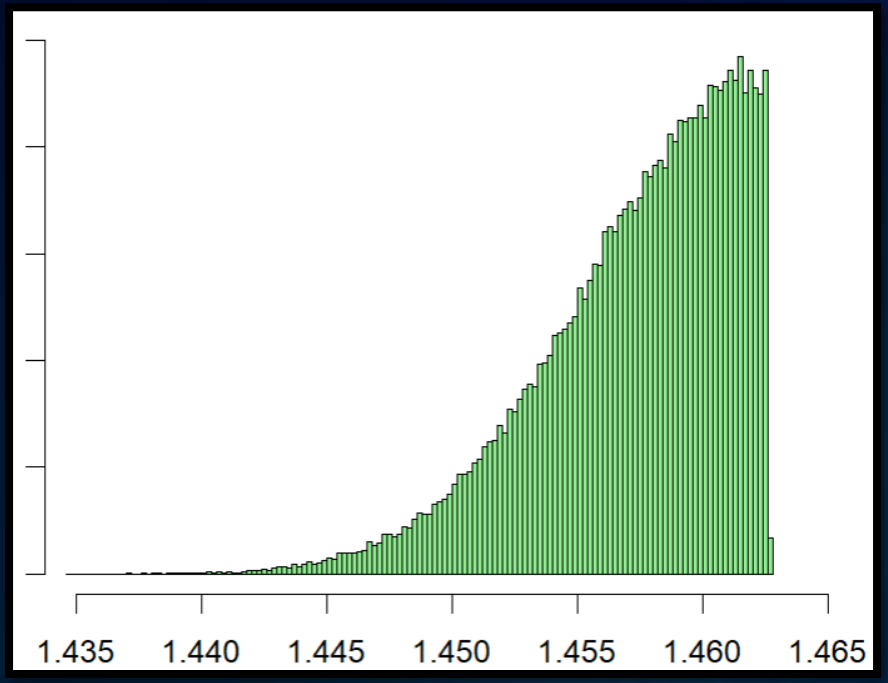
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for(k in 1:nmc){
  mcS = rnorm(26, 0, 1)
  mcstar = mystar(mcS, zmu0)

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}
# Run-time approximately 20 seconds

hist(mcstarstats, breaks=100, col='lightgreen', xlim=c(1.435, 1.465))
```

$$\star_{stat} := \frac{\frac{|\bar{x}|+1}{|\mu_0|+1}+3}{\frac{|\bar{x}|+1}{|\mu_0|+1}+2} \rightarrow_z \rightarrow \star_{z_{stat}} = \frac{\frac{0+1}{\left|\frac{\mu_0-\bar{x}}{s_x}\right|+1}+3}{\frac{0+1}{\left|\frac{\mu_0-\bar{x}}{s_x}\right|+1}+2} \rightarrow_{ideal} = \frac{4}{3}$$



Note that the empirical pdf does not contain the ideal value of $\star_{stat}$, so failing to reject H_0 when it is false (Type II error) is **really** large. But...

```
e_pval = mean(mcstarstats >= mystar(zS, zmu0))
e_pval # results in 0
```

So, we reject the claim that $\mu = 140$